Phishing Detective
An AI System That Scores Every Email

Logistic regression trained on 13,997 labelled emails, tested on 3,500 more, with a safety score for every message

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Google Colab notebook (shared link): https://colab.research.google.com/drive/1kLUIs3FJafw0vvu0TI55hswcTkTR9-u4?usp=sharing

The threat: phishing emails that steal credentials and money

191,561

phishing/spoofing complaints in 2025: the most-reported crime type

$3.05 billion

lost to business email compromise (BEC) in 2025

$20.9 billion

total cybercrime losses reported to the FBI in 2025, up 26%

- **What it is:** an email posing as a bank, a courier or your boss that asks you to click, log in or pay.
- **Impact:** direct **financial loss** (stolen accounts, wire fraud, gift cards) and a way in for ransomware.
- **Why AI:** nobody can inspect every message; an AI filter scores each one before it reaches the inbox.

Source: Federal Bureau of Investigation (FBI, 2026).

APPROACH · AI CONCEPTS

Chapter 5 concepts: machine learning + anomaly detection

1 Load

18,650 Kaggle emails: safe, or phishing and spam

›

2 Clean

Drop empty and duplicate emails, strip HTML → 17,497

›

3 Features

Words (TF-IDF), urgent words, personal-info requests, length

›

4 Train & test

80/20 split: 13,997 / 3,500

›

5 Score

Safety score 0–100 for any new email

MACHINE LEARNING: SUPERVISED CLASSIFICATION

- **Logistic regression** (main model) gives every word a weight, showing *why* an email is flagged.
- **Random forest** (300 decision trees) for comparison.

ANOMALY DETECTION: THE SAFETY NET

- A classifier knows only scams it has seen; anomaly detection flags **unusual** behaviour.
- *Example:* a first-time sender asking for a wire transfer → hold for review.

Sources: Parisi (2019, Ch. 3); Parisi (2019, Ch. 5); Cyber Cop (2023).

What phishing looks like compared with normal email



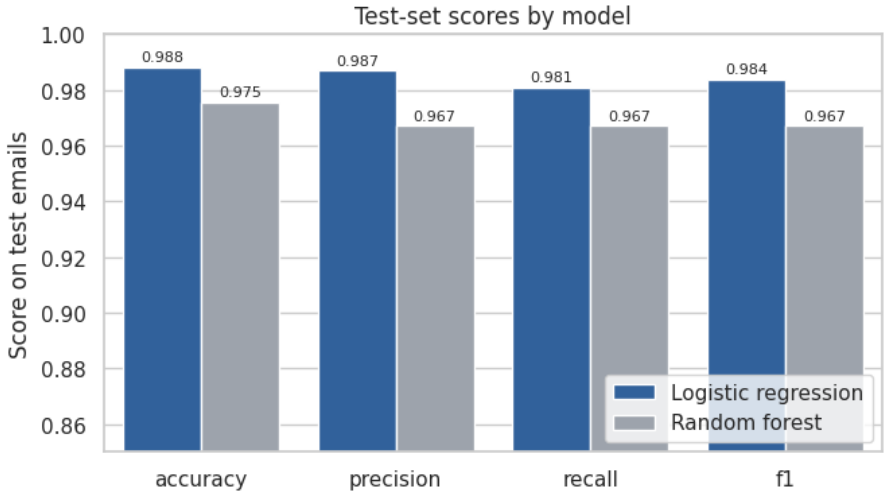
PER EMAIL, PHISHING HAS

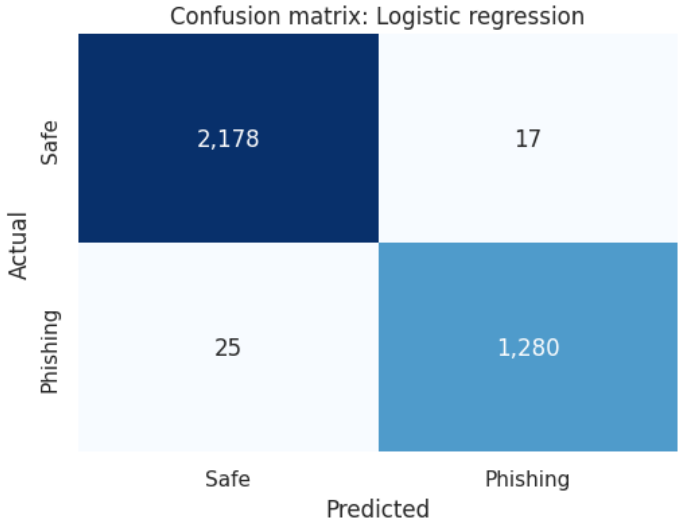
- **5.3×** as many exclamation marks
- **3.9×** as many money or prize terms
- **2.0×** as many requests for passwords, card or account numbers
- **1.5×** as many urgent words
- Links are **not** a giveaway: safe emails had slightly more.

Source: author's analysis of 17,497 cleaned emails from Cyber Cop (2023).

RESULTS · MODEL ACCURACY

98.8% accurate on 3,500 emails it had never seen





98.1%

of phishing caught (1,280 of 1,305)

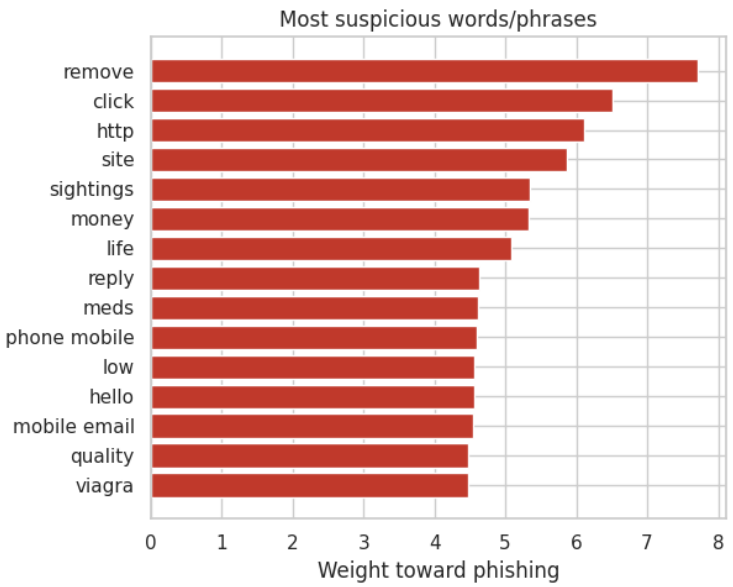
17 of 2,195

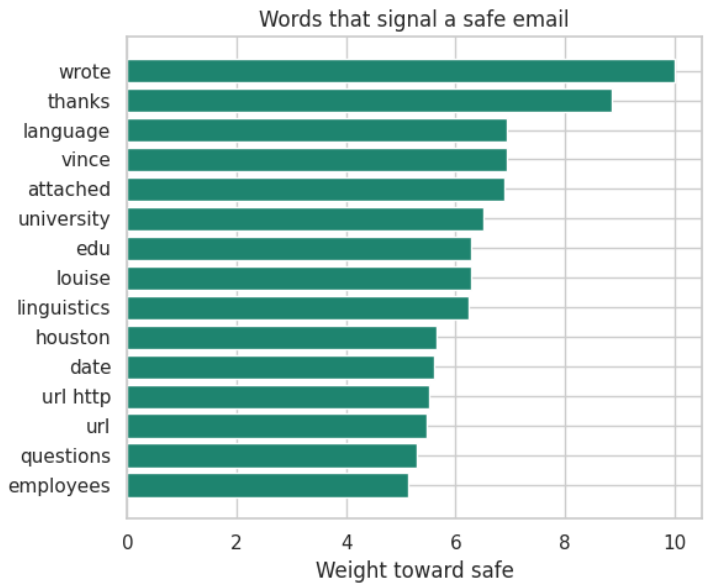
safe emails wrongly flagged

Logistic regression beat the random forest (97.5%), and its word weights show why it flags an email.

Source: author's analysis; test set = 20% of emails held out before training.

The most suspicious words and phrases the AI learned





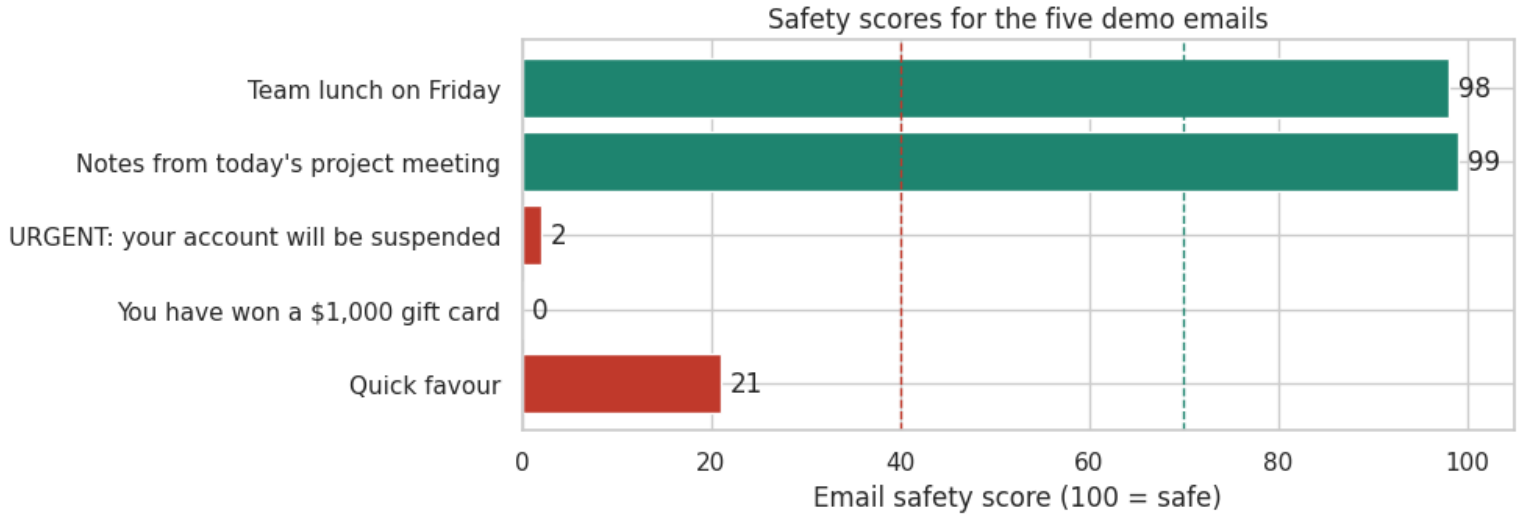
WHAT THE MODEL TREATS AS DANGEROUS

- **Top phishing words:** *remove*, *click*, *http*, *site*: links and calls to act.
- **Exclamation marks** and **money terms** also push toward phishing.
- **Caveat:** names and places (*vince*, *houston*) mark the safe archive, so accuracy holds for this dataset only.

Source: author's analysis; weights from the logistic regression.

SECURITY DEMO

Demo: email safety scores for five emails I wrote



SCORE = 100 × P(SAFE)

- **70–100** safe
- **40–69** be careful
- **0–39** likely phishing

All 5 verdicts matched my own label. The CEO-style *Quick favour* email has no link, yet scored **21**; top signals: length, account, bank, hi.

Source: author's analysis; demo emails written for this test, not taken from the dataset.

Trend: AI-written phishing, and how to defend against it

TREND: GENERATIVE AI IN PHISHING

In 2025, **22,364** complaints to the FBI mentioned AI, with **$893 million** in losses.

Chat generators “can quickly create official-sounding emails mimicking a company’s CEO” (FBI, 2026, p. 39); BEC scams involving AI cost businesses **over $30 million**.

The tell-tale spelling mistakes disappear.

HOW TO MITIGATE IT

- **Score wording, not spelling:** the link-free *Quick favour* scam still scored 21 from words such as *account* and *bank*.
- **Add anomaly detection** (Chapter 5): hold first-time senders who ask for payment.
- **Verify out of band:** confirm any money request by phone.
- **Retrain regularly** on recent, AI-written phishing.

Sources: FBI (2026); Parisi (2019, Ch. 5).

REPORT · WHAT I LEARNED

Security report and reflection

How accurate is it?	98.8% on unseen emails from this dataset: 98.1% of phishing caught, 17 false alarms in 2,195 safe emails.
What looks most dangerous?	Words such as remove, click, http and money, plus exclamation marks; phishing also asked for passwords or card numbers 2.0× as often.
How could it protect my family?	Score every email in the inbox and warn before anyone clicks or pays, especially on account-suspension and gift-card scams.

Reflection. Modules 3 and 4 applied AI to traffic and transactions; here the same workflow worked on language. I want to build email defences that pair models like this with anomaly detection and plain warnings my family can act on.

Full security report (100–150 words) in the notebook's Security report section.

APA 7TH EDITION

References

Cyber Cop. (2023). *Phishing email detection* [Data set]. Kaggle. https://www.kaggle.com/datasets/subhajournal/phishingemails

Federal Bureau of Investigation. (2026). *2025 IC3 annual report*. Internet Crime Complaint Center. https://www.ic3.gov/AnnualReport/Reports/2025_IC3Report.pdf

Parisi, A. (2019). *Hands-on artificial intelligence for cybersecurity: Implement smart AI systems for preventing cyber attacks and detecting threats and network anomalies*. Packt Publishing.

Notebook: Ruthvik_Nath_Bandari_Phishing_Detective.ipynb (Google Colab) · https://colab.research.google.com/drive/1kLUIs3FJafw0vvu0TI55hswcTkTR9-u4?usp=sharing